

Chapter 184

Immersive Audio Workflows

DaVinci Resolve offers substantial support for object and channel-based surround or immersive audio formats.

This chapter describes how to set up and mix with these formats when mixing in the Fairlight page.

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About Immersive Audio Formats

Surround audio formats use multiple audio channels to position sound sources around the listener to add a creative dimension to sound design. Their simplest formats, such as 5.1 and 7.1 surround, allow the mixer to send varying amounts of any track to a combination of speakers, placing the audio in various positions around an auditorium or living room, as centered in front, ambient from the rear, or weighted towards the left or right of the listener.

More sophisticated immersive formats such as Dolby Atmos™ or Ambisonics define a virtual soundstage where you can mix the resulting audio to match the speaker configuration or listening setup.

Dolby Atmos

The traditional surround sound experience outputs a specific set of monitoring channels that requires a specific number of monitor speakers placed in particular areas of a room, which can position sound approximately within a ring around the listener. Dolby Atmos improves upon this by being an object-based sound system operating within a 3D immersive space that can accommodate a wider variety of different speaker configurations using more speakers positioned around the listener. This increases dimensionality with more precise sound placement, adding height channels to produce sound that comes specifically from above. DaVinci Resolve supports Dolby Atmos output render formats 5.1.4 to 9.1.6.

A practical example of this difference can be heard when panning in a 7.1 mix; since you're sending signals to specific points in a speaker array, those points are fixed. Although the physical size of rooms can be larger or smaller, the mix is always sent to those assigned point speakers, so the experience from room to room may be inconsistent. By comparison, Dolby Atmos gives re-recording mixers a way to mix to an idealized space instead of fixed speaker positions. This means a Dolby Atmos mix, when played in a Dolby Atmos room, takes into account the actual dimensions of the space, as well as the number of speakers that are used, to recalculate audio playback to suit that exact space and playback

equipment, giving a more faithful recreation of the mix with much more specific sound placement, when necessary.

To give a clear example of the benefits of Dolby Atmos, think of a small theatre that has standard left, center, and right front-screen speakers. It then has four left surround and four right surround, four overhead left, and four overhead right. For this example, let's say you have sent a sound to the surround left in Atmos, positioned at about one-half of the distance from the screen.

Now move the Atmos mix to a much larger room with twice as many monitor speakers. The new theatre has eight left surrounds, eight right surrounds, eight overhead left, and eight overhead right. Playing in this new configuration, Atmos automatically calculates the ratio of the room and the new speaker array. In this example, when that sound is reproduced at one-half of the distance from the screen, Dolby Atmos calculates the ratio of the sound in relation to the new playback setup, so the listener hears exactly what the re-recording mixer intends.

For this example, let's say to have that audio playing back at one-half of the distance from the screen in the small room, the audio is played on the second surround speaker on the left. When this same audio is played in the larger room, Atmos determines that the fourth surround speaker is one-half the distance from the screen. The importance is not what speaker the sounds are assigned to but rather where in space the sound should be heard. Atmos calculates the ratio of the playback spaces and monitor speakers to faithfully reproduce a mix, rather than assign sounds to a fixed speaker position.

Dolby Atmos Speaker Configurations

It is important to understand that the term Dolby Atmos isn't restricted to describing any particular speaker layout. Dolby Atmos is a complex metadata-driven system that interprets the audio from a configured Atmos mix, determines the playback system of the end user, and calculates the mix to fit each particular space and system. Assuming the possession of a Dolby Atmos compatible system, if a 7.1.4 Atmos mix is played by someone with only a two channel playback system, then it will intelligently down mix the 7.1.4 Atmos to stereo. If a user has a standard 5.1 monitor system, then the 7.1.4 Atmos soundtrack will down mix to five channel surround with a subwoofer.

The naming of the channel configurations in the Dolby Atmos format includes the height channels in the nomenclature. Channel configurations are presented as three digits separated by periods, such as 7.1.4, which is a typical speaker configuration. The first digit describes the number of main, or ear-height monitoring channels that surround the listener. The second digit describes the number of subwoofer channels. The third digit describes the number of height channels, which are speakers positioned on, or in the case of a soundbar, pointed at the ceiling.

An example of a 7.1.4 Atmos monitor speaker configuration:

- Seven surround channels
 - Left
 - Center
 - Right
 - Left Surround
 - Right Surround
 - Left Back Surround
 - Left Right Surround
- One Subwoofer
- Four height channels

Enabling Dolby Atmos

Atmos must be enabled by going into Preferences > Video and Audio I/O > Immersive Audio panel and turning on Enable Dolby Atmos. When enabled, the Change Track Type contextual menu when right-clicking on a track will then include the additional Atmos track types.

The Components of a Dolby Atmos Mix

Mixes created in Atmos have several specifically identified components. These work together to create an immersive mix, but each element allows the Dolby Atmos system to fit the sound specifically to the space and speaker configuration during playback.

These consist of:

The Bed track: Contains the bulk of mixed audio, including dialog, ambient sound effects, and music. These sorts of sounds will contain panning information, but the panning will be general. Wind, traffic in the distance, room tones, and sync dialog would all most likely fall within the bounds of a standard bed track of 7.1.2 or 5.1.

Object tracks: Pinpoint the placement of sounds moving specifically across the immersive space. Users can use these specific tracks to create panning anywhere in the room. Sounds can fly through space, around a room from height channels, to side and back channels, mimicking the motion of objects on the screen. Object tracks use the Atmos metadata to do the calculations discussed in the last example. ADM files are Atmos exports used with the IMF file type as master deliverables, which are in the broadcast wave file format.

It's up to users to define which tracks are Bed tracks, and which are Object tracks; these track designations, are descriptions of what the tracks you create are going to be used for. Ultimately, it's up to the mixer which audio is organized on Bed tracks, and which is organized on Object tracks. Theoretically, users can create a mix consisting entirely of Object tracks if desired, but typically the mix would be split into beds consisting of generally panned sounds, and Object tracks for sounds requiring specific room placement.

Object tracks and height channels open up possibilities that were never possible prior to Atmos. Imagine a scene where a man is hiding from Police while a helicopter circles overhead, or a scene where kids are in a basement and are startled to hear loud footsteps above them. With Atmos, the audience can now experience these immersive sounds along with the characters.

These are real-world examples, but sound in animation, science fiction, and fantasy can explore space in ways that are only limited by the creator's imagination. Flying fairies or creatures can move about the space front to back, high and low. Perhaps there's a scene with a ship moving full speed ahead underwater, breaking up and out into the sky, flying and dodging weapons coming from all sides. Object tracks are ideal to pinpoint sound effects requiring spatial specificity, but more importantly, Atmos assures the re-recording mixer that the final choices made in the mix will be faithfully recreated from theatre to theatre, and from room to room.

NOTE: Buses feeding the Atmos master become beds, and Tracks feeding the Atmos master become objects.

Predetermined Dolby Atmos Master Rules

The first ten tracks in every Dolby Atmos Master are assigned as a bed by default. From there the default bed is a 7.1.2 bed, however, it can be designated to a 2.0 up to the 7.1.2 bed. There is no option to add objects to these first ten tracks. Starting at track eleven, the tracks that follow can then be made into various beds or objects as needed.

A simple way to think of this is that buses routed to the Atmos Master are treated as beds, carrying predefined multiple sources. An Atmos bed is a fixed surround format that can contain height channels. Objects, are generally mono but can send any format track that DaVinci Resolve supports and follow the metadata as a single object element. Objects routed to the Atmos Master are treated as tracks, carrying dynamic audio content and positional metadata. LFE will need to be sent to a bed, since objects are positional metadata. If the object is 5.1 for example, the LFE channel will need to be routed to a Bed bus in order to be rendered in the Atmos master.

The Dolby Renderer will render the file if it is dragged from the Media Pool into a track in the Timeline. In this case it will render the media with the embedded metadata to the master's output format. This is a simple way to monitor pre-mastered content and to perform simple actions, such as trimming or syncing for new packaging and deliverables.

When importing a Dolby Master file from the Fairlight menu, Fairlight > Immersive Audio > Import Master File, the resulting import will extract all of the audio and metadata for further content creation. This type of importing into Fairlight maps all of the metadata, tracks, beds, and objects from the master file into the Timeline, allowing you to adjust, process, and manipulate further, to rewrite panning, punch-in and add new media, and create a new Atmos Master.

Dolby Atmos files are a package of items. Simply linking files will not create an Atmos Master file. It is not possible, for instance, to take a rendered set of twelve tracks, link them, and then configure their outputs to a 7.1.4 Atmos bed. Note that 7.1.4 is not a Atmos bed type. Although this is the way typical PCM audio is linked and routed, it is not the case for Atmos content, which is far more than just a collection of tracks.

The renderer takes the 128 channels consisting of the beds and objects and renders it. Those channels are either internal sources or contained within the master file. A simple linked file will not play back through the Renderer. It must be played back as all linked files in Fairlight do, through the native monitor. The Dolby Atmos Renderer plays, renders, and extracts Atmos Master files. Fairlight allows for .atmos, ADM, and IMF file types to be imported and played through the Renderer.

NOTE: As mentioned above, the LFE can only be rendered in an Atmos master when routed as part of a bed due to the Dolby Atmos specification.

Dolby Atmos Integration

There is native Mac M1, Windows, and Linux support for the internal Dolby Atmos workflow including:

- Input
- Content creation
- Internal renderer
- Export of Dolby Atmos Master Files

There is a dedicated Dolby Atmos Master bus format which provides a master fader in the Mixer for the Dolby Atmos Renderer and is indicated by the Dolby logo in its header. It also provides a loudness history graph in its Timeline view with integrated, momentary, and short term loudness measures.

In the Bus Format window you will see that there are no outputs from this track; it is a control for the Dolby Atmos Renderer.

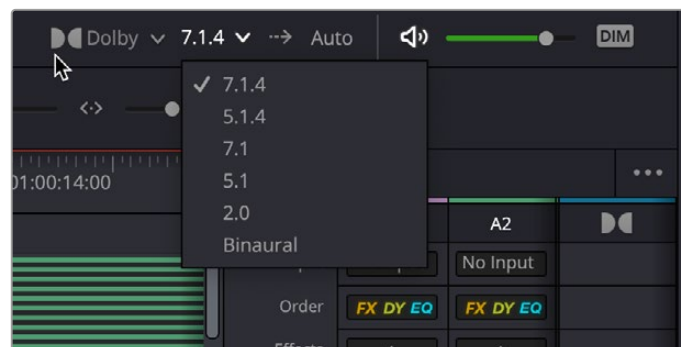
Only a single master bus can be created, and it will always mirror the format and metering of the internal renderer. When there are no patches to the Dolby Atmos Sends, all sources routed to a Dolby Atmos Master bus will target the master, as with a traditional bus workflow. In this send workflow, buses become beds and tracks become objects.

For Mac and Windows machines, the integration of the external Dolby Atmos Renderer has been updated, providing a much closer integration when running an external workflow. This has unified the tools available within the internal renderer workflows with the external renderer. This includes:

- External Renderer discovery within the preferences.
- UI enable to switch between the internal and external workflow.
- Tally of the master connection state and master file status.
- Storage and recall of the program metadata within the DaVinci Resolve Timeline, including all input configuration, trims, down-mix, and binaural settings.
- Direct control of the metadata from the renderer and binaural controls within DaVinci Resolve and bi-directional control of that program metadata from the server.
- Tally of the input configuration within the Patch page.
- Multi-client workflow support, allowing multiple instances of DaVinci Resolve to run attached to the same server for larger workflows.
- Tally of the active bed and object assignments on the Dolby Atmos Sends within the Patch page. When assigning a bed or track to the Atmos Master on the Patch page, beds are indicated with a purple marking and objects are indicated with a green marking.

Binaural Audio Monitoring

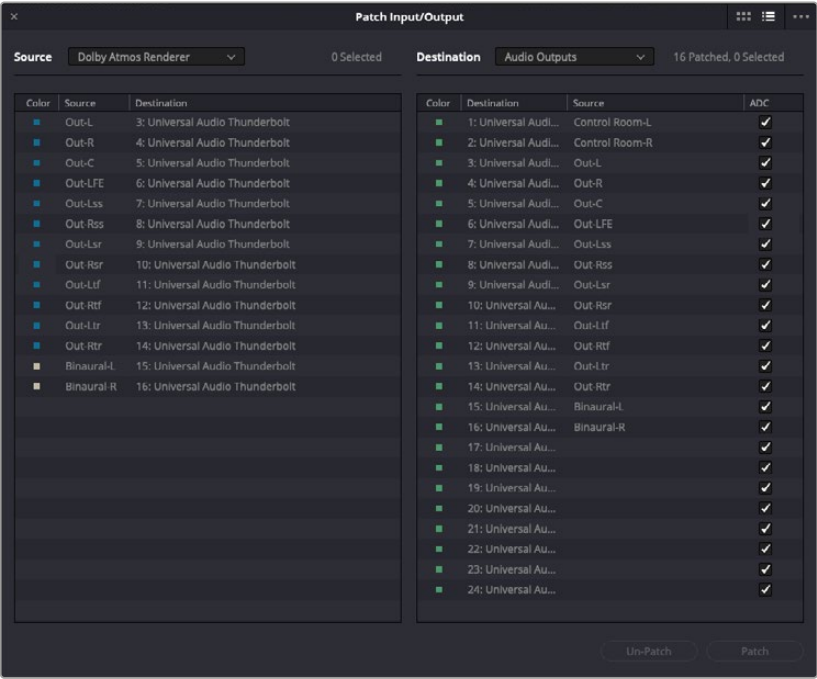
You can monitor your Dolby Atmos Master in Binaural by choosing Binaural in the monitor section.



The Binaural monitoring option

Depending on your monitoring system, you can also create a parallel speaker monitoring set up with Binaural. For instance, you can create a 5.1. external monitor mix along with a binaural headphone mix.

In Preferences > Video and Audio I/O, you can set Binaural to the headphone outs in the Audio I/O panel, and a 5.1 monitor in the Monitor Speaker configuration. Then, in the Patch Input/Output panel, you can route to the appropriate positional channels specific to your system and settings and also assign available channels for the headphones.



Using the Patch Input/Output to assign parallel binaural monitoring

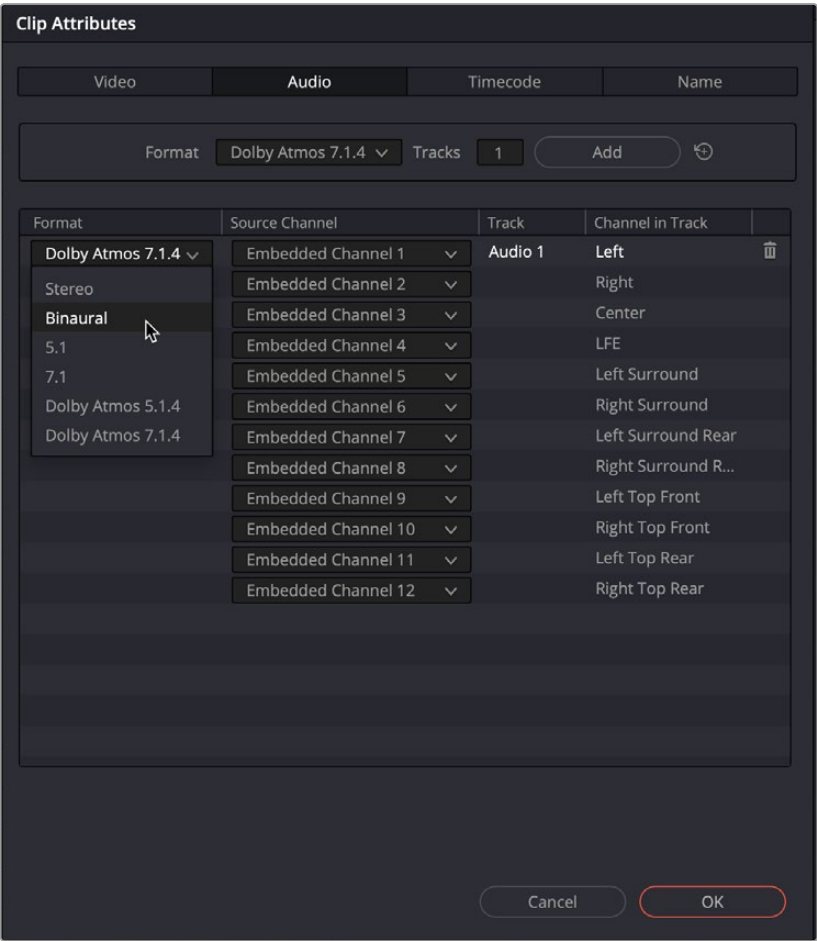
Binaural Render Options

Binaural rendering is a simulation of the distance between the sound source and the listener. In the Index there are options on how you can choose to render the binaural objects. The default option is Mid, but you can also choose Off, Near, or Far, depending on the desired rendering for binaural. These choices will all be stored in the Dolby Atmos file.

Edit Index Tracks Markers							
Tracklist							
	#	Name	Track Controls	Format	ADC	Group	Binaural
🔍	V1	Video 1					
🔍	A1	Composite Bed	🔒 R S M	7.1.2	✓	Comp B...	
🔍	A2	Dialog Object 1	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A3	Dialog Object 2	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A4	Dialog Object 3	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A5	Dialog Object 4	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A6	Dialog Object 5	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A7	Dialog Object 6	🔒 R S M	1.0	✓	Dialog	⇄ Near
🔍	A8	Music Object 1	🔒 R S M	1.0	✓	Music	⇄ Mid
🔍	A9	Music Object 2	🔒 R S M	1.0	✓	Music	⇄ Mid
🔍	A10	Music Object 3	🔒 R S M	1.0	✓	Music	⇄ Mid
🔍	A11	Music Object 4	🔒 R S M	1.0	✓	Music	⇄ Mid
🔍	A12	Music Object 5	🔒 R S M	1.0	✓	Music	⇄ Far
🔍	A13	Music Object 6	🔒 R S M	1.0	✓	Music	⇄ Far
🔍	A14	Music Object 7	🔒 R S M	1.0	✓	Music	⇄ Far
🔍	A15	Music Object 8	🔒 R S M	1.0	✓	Music	⇄ Far
🔍	A16	Music Object 9	🔒 R S M	1.0	✓	Music	⇄ Mid

Binaural choices in the Tracks Index for assigning perceived distance

You can use Clip Attributes to render a Dolby Atmos Master in Binaural. Simply right click the file to open Clip Attributes, select Binaural, and then right-click to create a new Timeline with the file. In the example below, a 7.1.4 file will be rendered to a Binaural audio file.



Using Clip Attributes to render to a Binaural file

Ambisonics

Ambisonics is a full-sphere multi-channel immersive “adaptive” audio format that lets you position sounds in the horizontal and vertical planes.

Compared to other surround formats, Ambisonics transmission channels aren’t speaker-specific and represent a complete sound field captured by a specialized microphone for an accurate positional depiction of “sound in space,” or generated from channel-based sources via sophisticated encoders.

The flexibility of this audio format makes Ambisonics compatible with various speaker configurations, such as 5.1 and 7.1.4, headphones, and large arrays in multiple applications, including theme parks, art installations, video games, and virtual reality.

You can even work and mix between channel-based and Ambisonics formats, without the need for third-party audio plugins.

DaVinci Resolve Fairlight’s full Ambisonics support includes spherical panning, Ambisonic bussing, Fairlight, AU, and VST Ambisonics effect plugins, accuracy up to fifth order High Order Ambisonics (HOA), and channel and binaural decoding to headphones.

Ambisonic Orders

Ambisonics “orders” define the mathematical precision and number of discrete channels to represent the sound field, ranging from the first to the seventh order. First order uses four discrete audio channels, while higher orders use more channels, requiring greater processing power.

Although DaVinci Resolve Fairlight supports up to fifth order Ambisonics, requiring 36 channels, you can get great results using third order and 16 channels.

Ambisonic File Support

DaVinci Resolve supports Ambisonic files in WAV, BWAV, or .caf (including 10A AMB and AmbiX) formats.

Adding a Media Pool clip to the timeline or creating a new timeline that refers to a clip creates a track of the appropriate type, matching the file’s Ambisonic order and channels.

Enabling Ambisonics

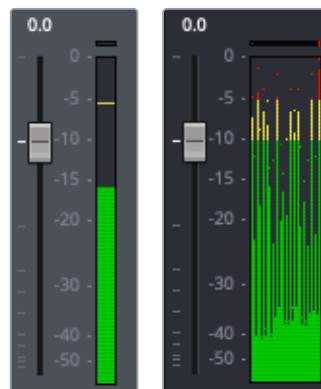
Ambisonics is enabled by going to Preferences > Video and Audio I/O and clicking the Enable Ambisonics checkbox in the Immersive Audio section.

Ambisonic Metering Options

Mixer and Track Header Meters

Fairlight Ambisonic metering utilizes a single, composite bar-graph meter in the Mixer, Track header, and Meter panel. Unlike other audio formats, they don’t correlate to specific speaker positions or levels we would track visually to determine what is happening with the signal.

Alternatively, you may want to use bar-graph meters that display Ambisonic signals as discrete, separate bus streams by choosing Fairlight > Immersive Audio > Ambisonics Channel Metering.



Default Ambisonics metering (left) and Discrete Channel metering (right)

Ambisonics Metering

This Fairlight FX audio plugin offers two different visual metering options for analyzing your audio. The Ambisonics meter can be added to an audio track or bus via the channel strip’s Effects section.

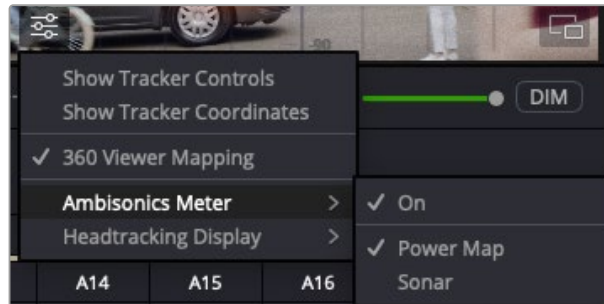
For more information about the Ambisonics Meter, see *Chapter 181, “Fairlight FX.”*

Video Viewer Overlay

The Fairlight page also offers Ambisonic metering options as overlays on the Video viewer, allowing you to analyze the sound intensity against the picture.

To enable the Ambisonics Meter overlay:

- Activate 360 Viewer Mapping in the Video Viewer tools menu.
- Go into the Ambisonics Meter submenu, click On, and select one of the metering modes.



Enabling the Ambisonics Meter overlay

Ambisonic Effects

While most Fairlight FX plugins, such as the Multiband Compressor, Limiter, Reverb, and Channel Dynamics and EQ, are Ambisonic-native, DaVinci Resolve also supports compatible Ambisonic-native third-party AU and VST effects.

For more information on the Channel Dynamics and EQ plugins, see *Chapter 177, "Mixing in the Fairlight Page."*

For more information on Audio Effects plugins and how they can be used in DaVinci Resolve, see *Chapter 180, "Audio Effects."*

For more information on individual Fairlight FX plugins, see *Chapter 181, "Fairlight FX."*

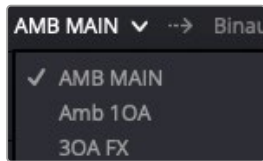
Setting Up an Ambisonic Mix

To organize and set up your Ambisonic mix:

- The first thing you need to do is determine which Ambisonic order you want to work in. A higher order such as third order (16 channels) is best for precise positioning.
- Create Ambisonic busses and assign your source tracks or other busses to them. You can assign Ambisonic busses to other Ambisonic busses, even with differing orders. The Fairlight engine and FlexBus will automatically handle any required up- or down-mixing.
- Use the 2D or 3D Spherical Panners on each channel to position your sources in the Ambisonic space. You can use any channel-based format from mono to stereo to, for example, 7.1.2, and position them freely in the spherical sound field.

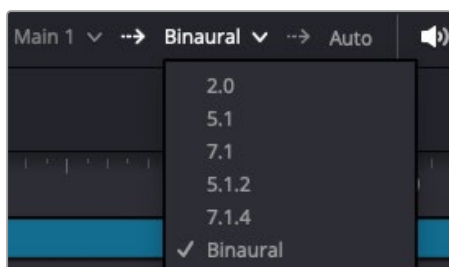
For more information on the 2D and 3D Panners, see *Chapter 177, "Mixing in the Fairlight Page."*

- Click the Bus Monitoring menu to the far right of the Transport control, and select the source you want to monitor.



Monitoring Source selection

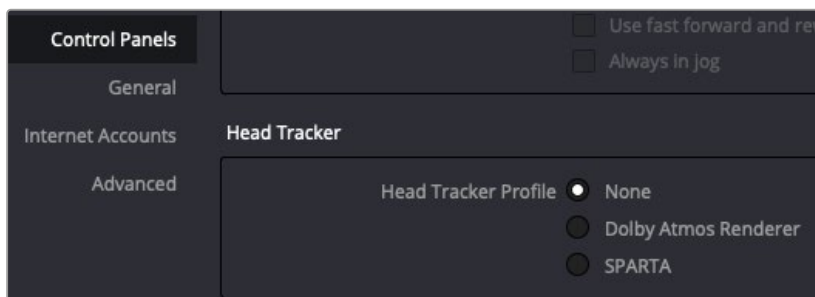
- Choose your output format in the Monitor section dropdown next to the speaker output level control. You'll notice various options, including Binaural for monitoring on headphones. Once you've selected a format, the corresponding decoder is automatically placed in the monitoring path, and the audio is routed to your Audio I/O and sent out to your speaker system.



Output Format selection

Binaural Monitoring with Head Tracking

If you've selected the Binaural option as your output format, you can also monitor your Ambisonics mix on headphones with support for Head Tracking for an enhanced monitoring experience.



Head Tracker preferences

Enabling Head Tracking

Head tracking functionality can be activated by going to Preferences > Control Panels > Head Tracker. This adds two buttons to the left of the Monitor Bus Source dropdown immediately below the docked Video Monitor panel:

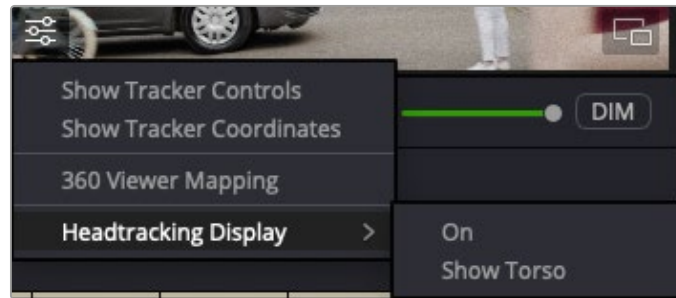
HT: Switches Head Tracking on and off.

CAL: Calibrates the current tracker direction as the "Front" or point of origin.

Head Tracking Video Viewer Overlay

Binaural Head Tracking also lets you add an overlay to the video viewer for analyzing your mix against the picture, by clicking Headtracking Display > On, which adds an onscreen crosshair, indicating the current head (headphone) orientation.

This submenu also includes a Show Torso option which adds a head and torso to the video viewer, that rotates as the head tracking angles and positions change.



Head Tracking Video Viewer overlay

Parallel Ambisonic and Channel-based Workflows

You can create Ambisonic and channel-based mixes in the same timeline and simultaneously render multiple outputs in different formats.

Because of the common panning data, it's also possible to set up a parallel mix in an Ambisonic timeline by converting an existing production of a differing surround format (e.g., Dolby Atmos) to Ambisonics format and importing that into your production.

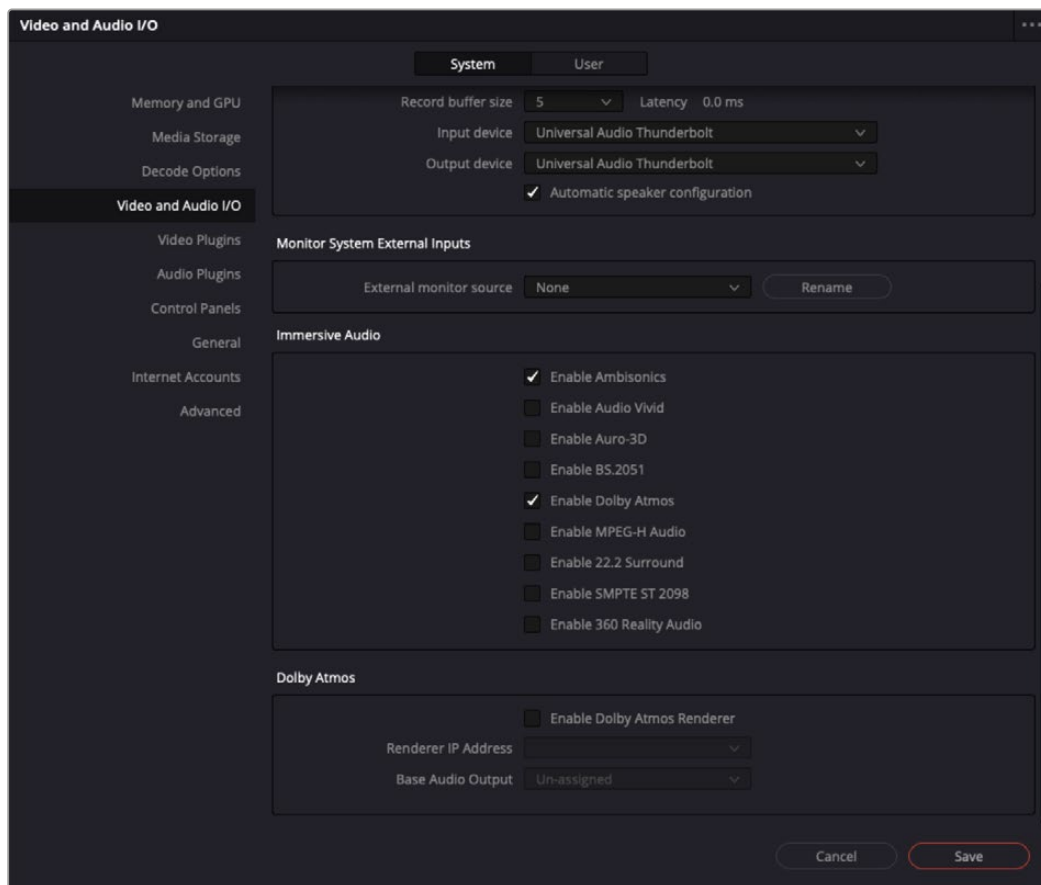
Ambisonic Rendering

Ambisonic mixes can be rendered on the Deliver page as either a multichannel adaptive file with the number of streams determined based on the Ambisonic order of the mix (e.g., 1OA, 2OA, etc.) or downmixed to the format of your choice, such as 7.1.4.

Immersive Format Configuration

In Preferences > Video and Audio I/O > Immersive Audio, you can enable the various Immersive options that are available. These are Ambisonics, Audio Vivid, Auro 3D, BS.2051, Dolby Atmos, MPEG-H Audio, 22.2 Surround, SMPTE ST 2098, and 360 Reality Audio.

In Preferences > Video and Audio I/O > Dolby Atmos, you can configure the use of an external Dolby Atmos Renderer for Dolby Atmos monitoring and mastering. You can manually enter the IP address of the server, or select a discovered server using the dropdown menu. You should also choose the base audio output of the audio outputs, which will be used to send the audio to the external renderer.



Controls for enabling immersive audio formats in the Video and Audio I/O preferences

Exporting ADM BWF

You can export a Dolby Atmos master file as an audio-only ADM BWF, right from the Fairlight timeline. These same options are also available in the Deliver page. Exporting a Dolby Atmos master file from the Fairlight timeline uses the Timeline name as the filename. Be sure to change the Timeline name to the desired filename.

- In the Media Pool, locate the current Timeline.
- Change the Timeline name to your desired export filename.
- As with all other bouncing and delivery methods, you will need to mark a range in the Timeline to export.
- Press R for the Range Selection tool. Double-click any of the Timeline clips to set a range for the entire clip.
- Choose Fairlight > Immersive Audio > Export Master File.

The screenshot shows the 'Export Immersive Master' dialog box with the following settings:

- File Name: Timeline Name
- Tag: (empty)
- Format: Dolby Atmos ADM BWF
- Sample Rate: 48000 (Delivery)
- Source: 48000 (Delivery)
- FFOA: 96000 (Archive)

Export Immersive Master options

- In the Export Immersive Master dialog, set the File Name to Timeline Name and the Format to Dolby Atmos ADM BWF. You can select either 48000 (delivery) or 96000 (delivery) for the Sample Rate of the exported file. The Source is automatically set to the Atmos Send Patching. This patching passes the signal through the sends to the internal Dolby Atmos Renderer for processing and to generate a new Dolby Atmos master file.
- Click Export.
- In the Export Immersive Master finder window, navigate to the folder you want file to save to. Click Save.

The screenshot shows the 'Export Immersive Master' dialog box with the following settings:

- File Name: Timeline Name
- Tag: (empty)
- Format: Dolby Atmos ADM BWF
- Source: Atmos Send Patching
- FFOA: 01:00:00:00

The 'Export' button is highlighted with a red border.

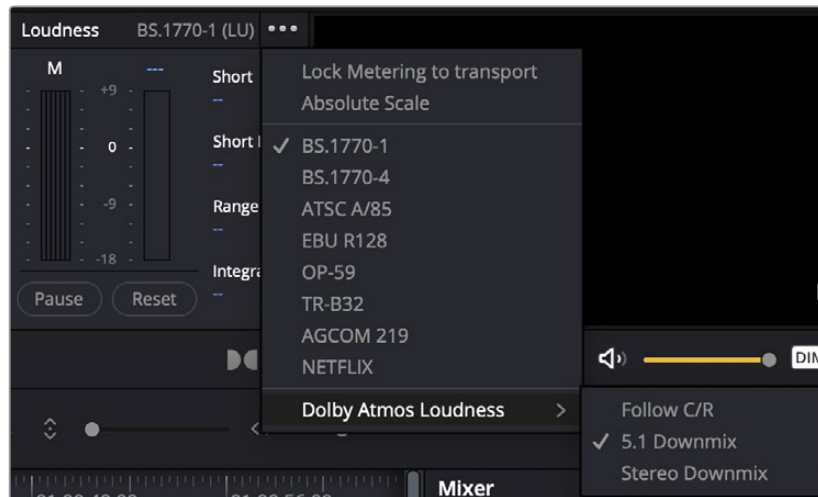
The Export Immersive Master dialog shows the source and format for the new ADM export.

The Export allows for several types of file types:

- Dolby Atmos ADM BWF
- Dolby Atmos IMF IAB
- Fraunhofer MPEG-H Production
- Fraunhofer MPEG-H Production XML

Parallel Loudness Measurement

A parallel 5.1 down-mix can be performed during the render as a source for a 5.1 loudness measurement. Certain delivery specifications (such as for Netflix) require that the loudness measurement of Atmos Masters be performed at 5.1. You can enable this parallel 5.1 (or stereo) render for the loudness measurement and monitor in the room's native format.



Selecting a Downmix option

Low-latency Rendering

Previous versions of the Dolby Atmos Renderer were able to run at a specific latency of 10.67 ms. This restriction has been removed, allowing selection of lower latency engine processing block sizes, and thus lower monitoring latency through the Dolby Atmos renderer. However, the larger block size is enforced by the renderer during certain operations, such as Binaural renderer, or when a parallel loudness down-mix is enabled.

Surround Buses in Fairlight (Studio Version Only)

The FlexBus structure in Fairlight enables user-definable buses. Atmos mixes call for several bus formats that are available in DaVinci Resolve 17 and later.

- 9.1.6
- 22.2

Object-Based Format Support (Studio Version Only)

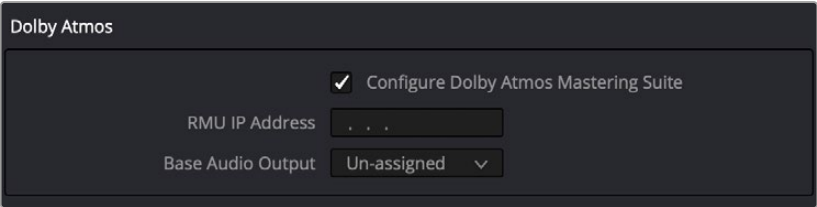
- Dolby Atmos with support for 7.1.2 and 7.1.4
- MPEG-H with support for 5.1.4, 7.1.4, and 7.2.3
- SMPTE ST.2098 with support for 9.1 OH, 9.1 HT, 11.1 HT, 13.1 HT, and 15.1 HT

Auro-3D Support (Studio Version Only)

- Auro-3D with support for 9.1, 10.1, 11.1 (7+4), 13.1

Dolby Atmos Configuration Controls

The Video and Audio I/O panel of the Resolve System Preferences lets you enable and configure the use of a Dolby RMU for doing Dolby Atmos mixing. You can enter the IP address of the RMU, and choose the base audio output.

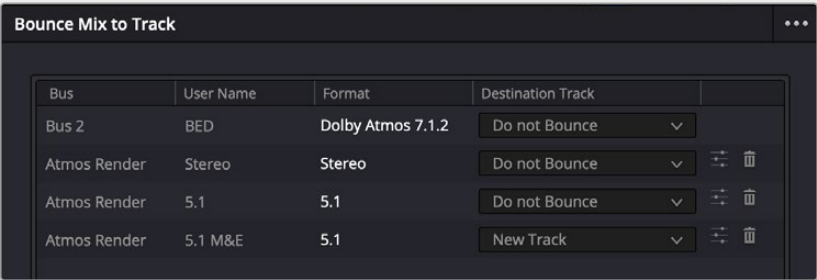


Configuring Dolby Atmos in the Resolve System Preferences

Atmos Re-Renders

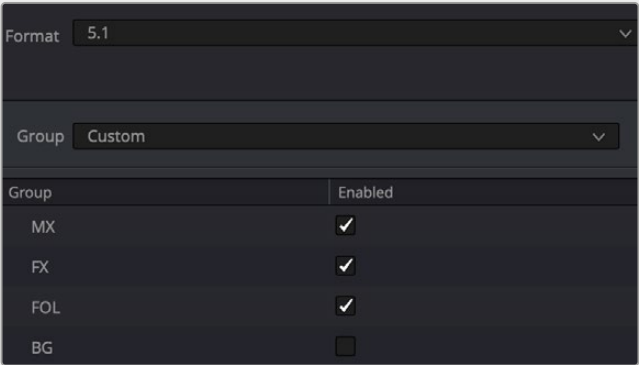
In order to provide parity with the external renderer, you can create presets that let you define a PCM output format and a set of sources using the bed, object, and group definitions (such as 5.1).

Bounce Mix to Track accommodate these re-render presets in Dolby Atmos timelines, allowing you to generate multiple deliverables in a single pass. These render presets are stored and recalled as part of the timeline.



Bounce Mix to Track Dolby Atmos re-render options

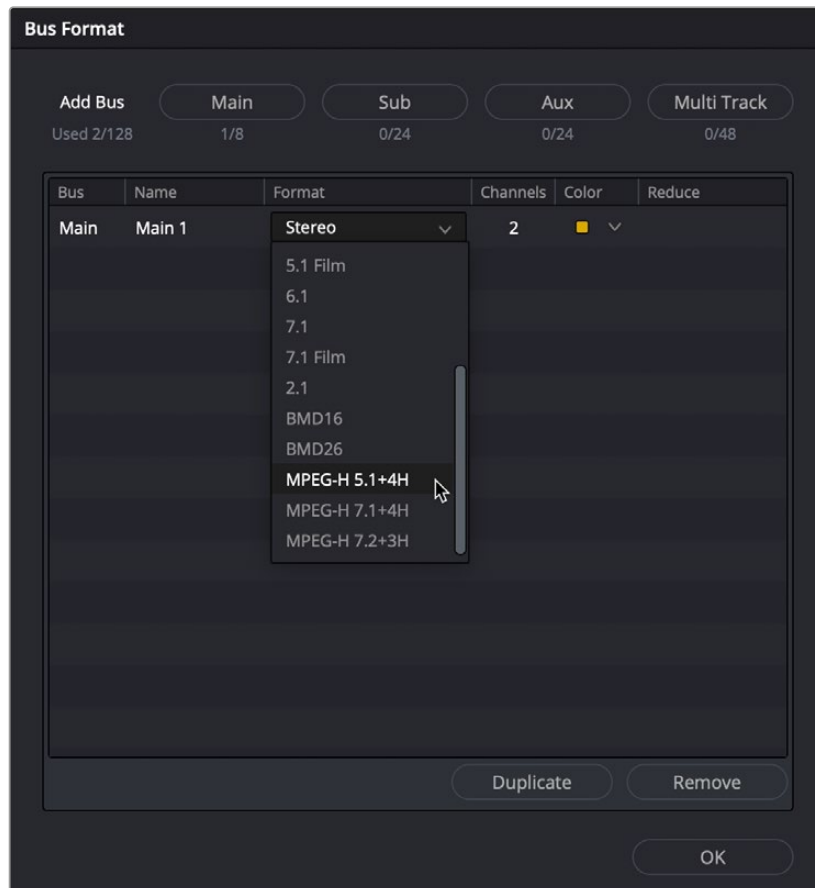
In addition to render format, you can select the source from a combination of beds, objects, or defined VCA groups.



Dolby Atmos Render Preset options

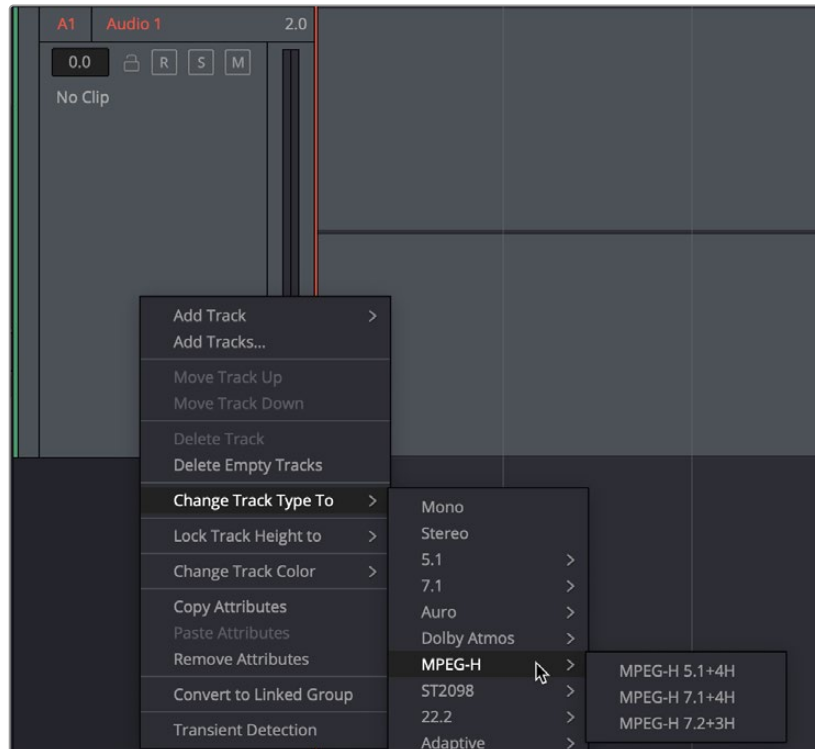
MPEG-H Authoring

DaVinci Resolve enables MPEG-H authoring. This includes Native MPEG-H track and bus formats and monitoring, including the ability to define basic track-level meta-data for export into an MPEG- H scene, and export of a MPEG-H Master file. Once enabled, the formats become available for selection as a bus, track, or monitoring format, in the Bus Format window.



The MPEG-H format options in the Bus Format window

These formats also become available for multi-channel track assignment.



The MPEG-H format options in the Track Format submenu

Tracks are mixed natively in a similar manner to most immersive content, with the creation of a bed mix consisting of a set of immersive object tracks that use dynamic panning. These are combined onto the main bus to form the immersive mix.

Track Configuration

In addition to this process, once the format is enabled, a set of MPEG-H meta-data columns become enabled in the DaVinci Resolve Track index, including Track Type, Kind, Language, SW Group, and Preset.

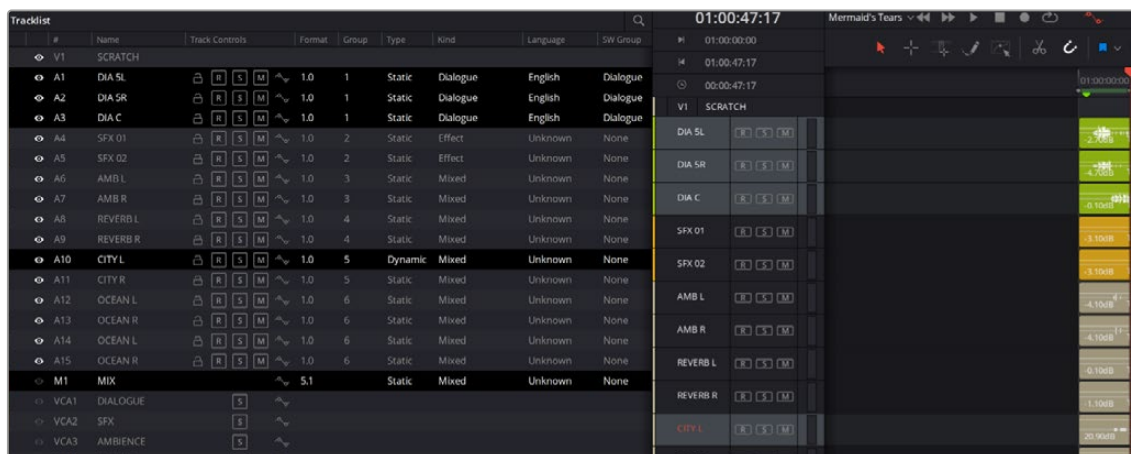
For more information about these columns and how to configure them, see *Chapter 170, "Using the Fairlight Page."*

Edit Index Tracks Markers										
Tracklist										
	#	Name	Track Controls	Format	Group	Type	Kind	Language	SW Group	Presets
👁	A1	VO	🔒 R S M	1.0	0	Static	Undefined	Unknown	None	Default
👁	A2	AMB 01	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default
👁	A3	AMB 02	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default
👁	A4	AMB 03	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default
👁	A5	SFX 01	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default
👁	A6	SFX 02	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default
👁	A7	Music	🔒 R S M	2.0	0	Static	Undefined	Unknown	None	Default

When MPEG-H is enabled in the Project Settings, the Tracks panel shows additional columns of information for defining each track in the Timeline

Export

Once everything is configured, and your project is mixed, you export a master file. In order to do this, you must select a range In and Out point on the Timeline to define an export range. Additionally, you must define whichever busses are designated for rendering by defining their Kind, and you can select tracks you want to export additionally.



Selecting tracks and busses to export

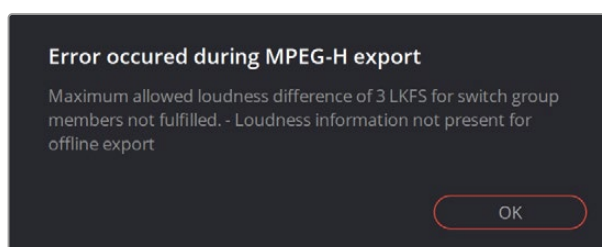
NOTE: An MPEG-H master file can only contain a maximum of fifteen channels total. If the selected track and bus stems exceeds this, then the export will fail, issuing a warning. The same will occur if a range is not selected.

To export an MPEG-H mix:

- 1 Choose Fairlight > Immersive Audio > Generate MPEG-H Audio File.
- 2 Choose a save location and name, and click Save.

At this point, all defined buses are rendered. Then, track loudness is measured for compliance. Lastly, the source audio is exported and the metadata embedded into the deliverable MPEG-H wav file.

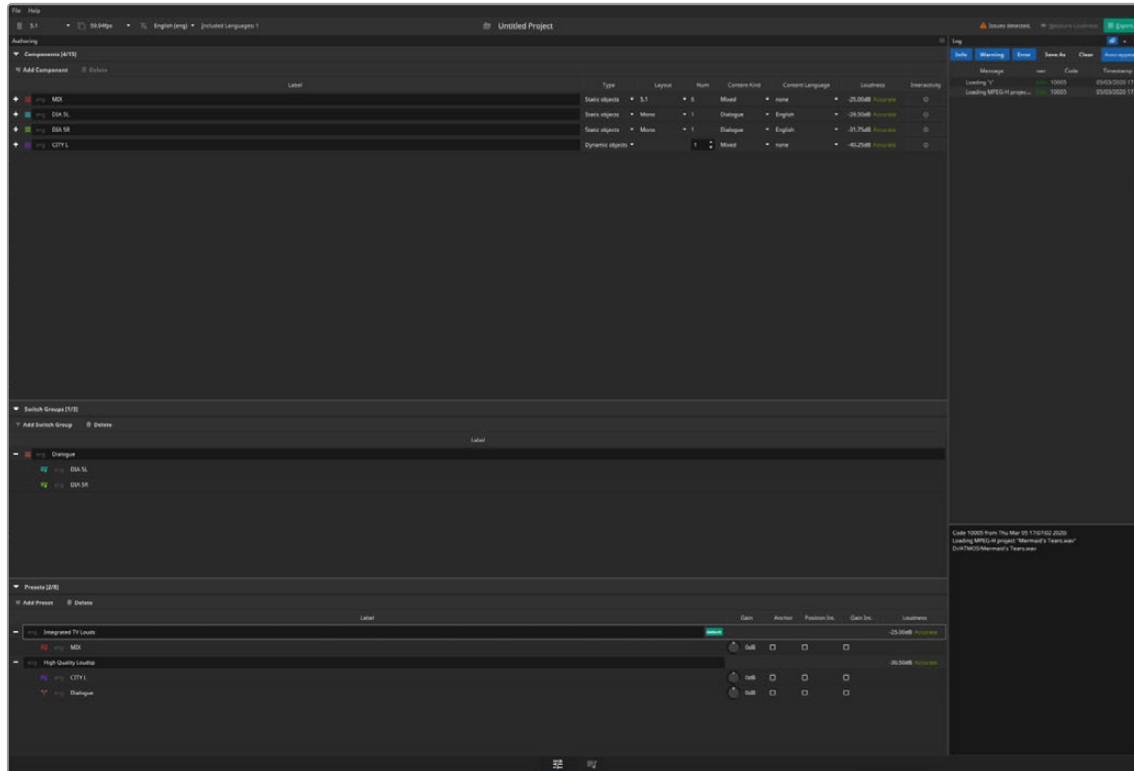
There are several error conditions, which are deleted during the export process, that will all cause the export to fail. For example, all tracks must contain audio. Audio within switch groups must be within a specific loudness tolerance of one another. If any of these conditions occur, a dialog will appear.



Warning dialog during MPEG-H export

Quality Control

You can do a quality control check of the final audio export using a Fraunhofer MHAPl tool which can open up the exported content. This includes compliancy tests, and visualization of the contained metadata. It can also render and monitor the content, but this aspect of the tool is basic, and only works with external audio that's linked to the metadata. In other words, you must manually link the components back to the source audio files.



Quality control using the Fraunhofer MHAPI tool

B-Chain Support for Audio Monitoring (Studio Version Only)

Choosing Fairlight > Immersive Audio > B-Chain Control opens the B-Chain Control window. Using traditional cinema audio postproduction terminology, the “A-Chain” is all of the bussing and signal processing that happens within the mix (in our case using the Fairlight page) and the “B-Chain” is the signal processing, amplification and speaker system that takes the sound from your workstation’s output and gets it “into your ears.” This typically includes all necessary signal decoding or pre-processing hardware, amplification systems, and speaker setups for most professional and commercial listening environments, especially for immersive audio or surround sound formats. These options are complex and have many variables in terms of speaker configuration, selection, and placement, output channel timing, attenuation, and phase, and a host of other considerations.

In DaVinci Resolve, these B-Chain controls affect how the channels of audio being output by the Main you're mixing is mapped to the actual audio signals being output from your workstation into the amplification and speaker system of the current viewing environment. The extensive level of control the B-Chain presets offer is most useful in situations such as surround-sound configured grading and mixing theaters, to define how the immersive audio standard you're mixing is mapped to

the speaker setup of your particular application, and to fine-tune the channels being output to each speaker specifically for your environment. In simpler monitoring situations, the B-Chain controls can be useful for configuring the standard you're mixing with to play out of a non-standard speaker setup in your room.

Ultimately, the B-Chain controls are designed and intended for users who need to create custom, fine-tuned settings for their specific monitoring situation. For this reason, pretty much every real-world use of these B-Chain controls will require custom configuration for your unique environment.

Overview of Setting Up a B-Chain Configuration

Setting up a B-Chain configuration takes some doing, but once set up, you have a preset that's easy to reopen at any time. In the following example, a B-Chain preset will be configured to convert a 5.1 surround output to a 3-channel left-center-right room speaker setup.

To set up a B-Chain:

- 1 Choose Fairlight > Immersive Audio > B-Chain Control to open the B-Chain Control dialog. This exposes the monitoring controls but not the configuration controls.
- 2 Click the Option menu in this window and choose Setup B-Chain Presets. The B-Chain Setup window appears.
- 3 Click the New button to create a new B-Chain preset, then enter a name in the dialog and click OK. In this example we'll use the name "5.1 to LCR." The Name you enter here is the name of the saved setting data.

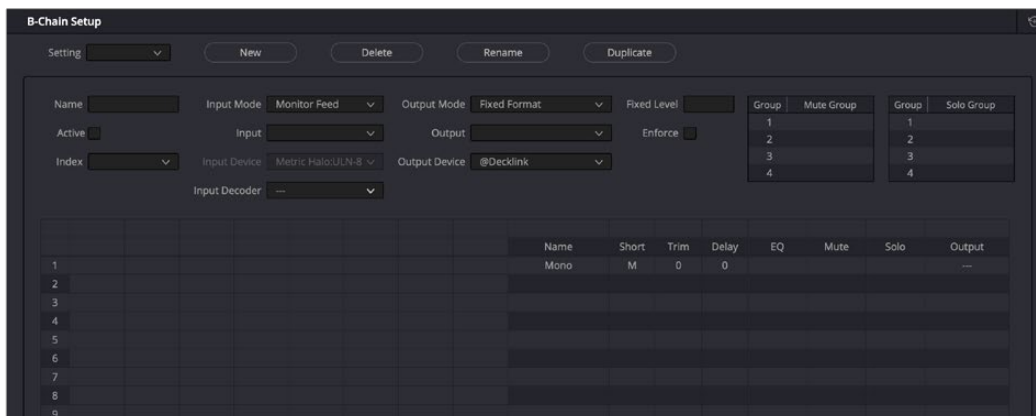
Additional buttons let you Delete, Rename, or Duplicate presets that you've already created once you choose them from the Setting pop-up.

- 4 Enter a label in the Name field (this is the name that appears in the preset menu). The name you enter here is the name that appears in whatever pop-up menus you select these options from.
- 5 Choose Input Mode > Fixed Format, and Input > 5.1 from the pop-up menus. These define the signal coming out of your mix. (Input Mode > Monitor Feed sets the B-Chain to listen to the selected monitor source, while Input Mode > Custom Format lets you choose an arbitrary number of channels.)
- 6 Choose an Output Mode > Fixed Format, and Output > LCR from the next set of pop-up menus. These define the speaker setup you'll be playing the mix out of. In this example, we've chosen LCR, which stands for Left, Center, Right.

At this point, you can see that once you've defined the Input and Output channels, the "Inputs/Outputs" table of cells becomes populated:

- At the left of this table, each channel of your selected Input occupies a vertical column. In this example, the Input columns are labeled L, R, C, LFE, Ls, Rs, corresponding to the 5.1 channels you've selected to be output. It's possible to specify setups with many more channels, in which case the left half of the table becomes scrollable.
- At the right of this table, each channel of your selected Output occupies a horizontal row that intersects the Input columns. In this example, the rows are labeled Left, Center, Right.

In this way, every Input column intersects with each Output row, making it possible for you to assign how much of each Input is fed to each Output by entering values into each intersecting cell.



The Inputs/Outputs table of the B-Chain preset window

- 7 To assign Input channels to Output channels, double-click the field where the desired Input intersects the desired Output, and type a value to set how much level from the Input to assign to the Output. Typing 0 assigns Input to Output at “unity” gain (0dB). Numbers you type here specify tenths of a dB. Typing -3 assigns half of the level from Input to Output. Typing -100 mutes that Input to that Output completely. Blank cells with no value make no assignment. In this example, we’re typing:
 - a) 0 into cell L-1 to assign all of the Left source to the Left output
 - b) 0 into R-3 to assign all of the Right source to the Right output
 - c) 0 into C-2 to assign all of the Center source to the Center output
 - d) -6 into Ls-1 to assign a fraction of Left surround to the Left output
 - e) -6 into Rs-3 to assign a fraction of Right surround to the Right output
 - f) -8 into LFE-1 to assign a smaller fraction of LFE to the Left output (LFE is not directional)
 - g) 8 into LFE-3 to assign a smaller fraction of LFE to the Right output as well (LFE is not directional)

At this point, we’ve fully defined how much of which Input channels go to which Output channels. In the process, you can see that this table interface makes it possible to assign the full levels of Input channels to Output channels, to assign partial levels of Input channels to Output channels, and to assign fractions of Input channels to multiple Output channels.

	L	R	C	LFE	LS	RS			Name	Short	Trim	Delay	EQ	Mute	Solo	Output
1	0			-8	-6				Left	L	0	0				
2			0						Centre	C	0	0				
3		0		-8		-6			Right	R	0	0				

The Inputs/Outputs table of the B-Chain preset window fully populated to assign inputs to outputs

- 8 At this point, you should notice there are additional columns to the right of the Output channels that provide various options that let you fine-tune each channel’s output in order to optimize the acoustics and layout of your room:
 - Trim each channel (in tenths of a dB)
 - Add a Delay (in milliseconds)
 - Add specific EQ (None, LFE Only, Surr Mode)
 - Mute or Solo groups.

- 9 The last thing you need to do is to right-click each Output channel row's right-most Output column, and choose the hardware output of your audio or video+audio interface that you want that channel to go to. See one of the following I/O Setup procedure for how to rename these options to make this easier.

Name	Short	Trim	Delay	EQ	Mute	Solo	Output
Left	L	0	0				1: Decklink
Centre	C	0	0				2: Decklink
Right	R	0	0				3: Decklink

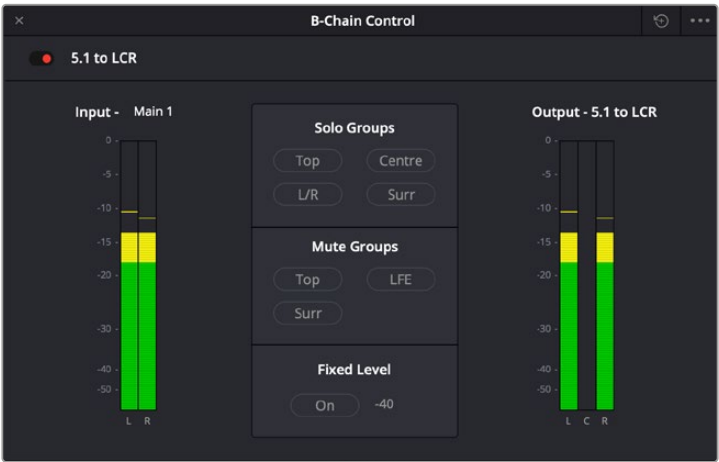
Assigning I/O channels to output to

Once you've finished setting up your B-Chain preset, click OK, and that preset will be saved and available for use.

To choose a B-Chain preset to use and enable B-Chain:

- 1 Choose Fairlight > Immersive Audio > B-Chain Control to open the B-Chain Control dialog.
- 2 Turn on the toggle at the upper left-hand corner of the window to enable B-Chain, and choose a preset from the pop-up menu.
- 3 Choose the Input you want to assign that B-Chain preset to.
- 4 A set of Solo and Mute buttons lets you selectively choose sets of channels you want to listen to in isolation, or mute from the whole, in order to better evaluate selected portions of your mix.
- 5 A Fixed Level button lets you fix the output level in the DaVinci Resolve UI to a single value. With this enabled, you cannot drag the level slider to change the monitoring volume.
- 6 When you're finished using B-Chain for monitoring, turn off the toggle at the upper left-hand corner.

NOTE: While the B-Chain is enabled, the Speaker Setup controls in the Video and Audio I/O panel of the Resolve System Preferences is disabled.



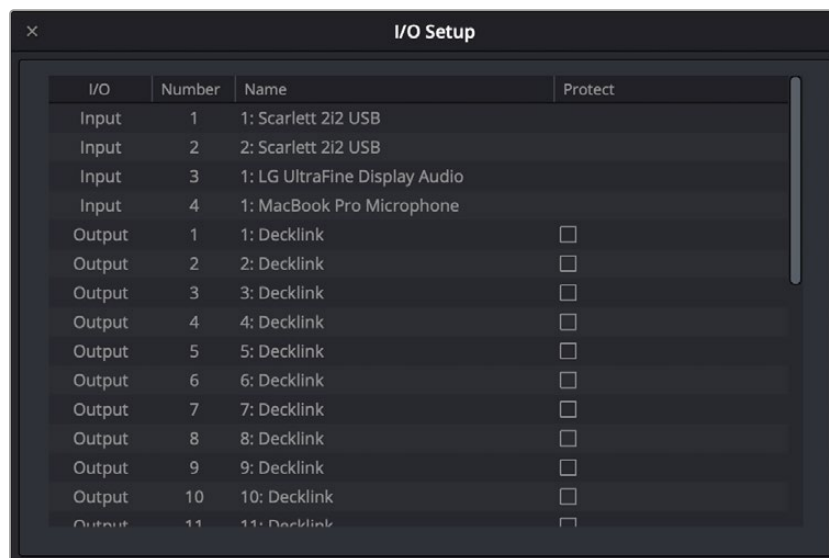
The B-Chain Control Window where you can choose and enable a B-Chain preset to affect the output

To name the various outputs of your system's channels for easy assignment:

- 1 Choose Fairlight > Immersive Audio > B-Chain Control to open the B-Chain Control dialog.
- 2 Click the Option menu in this window and choose Setup I/O Names. The I/O setup window appears, showing every single audio input and output on every audio interface that's connected to and recognized by your system.

The I/O Setup window lets you rename the Inputs and Outputs associated with your workstation setup, to make it easier to configure Fairlight when you're managing hundreds of channels. The names you choose here are shown everywhere in DaVinci Resolve where channels are selectable and/or exposed.

- 3 To rename a channel, double-click that channel's field in the Name column, type a new name, and press Return.
- 4 To protect a channel, click the checkbox in the Protect column. Protected channels are prevented from being patched to. This is intended for potentially high-volume speaker outputs, to which it would be hazardous to accidentally connect the wrong input. Protected channels can be seen everywhere patches can be made, but they're grayed out. Protected channels can only be used in the Speaker Setup controls in the Video and Audio I/O panel of the Resolve System Preferences, and in the B-Chain Setup window.
- 5 When you're finished, close the window. The names and protected status of channels are a system-wide setup.



The I/O Setup window where you can rename your system's audio outputs and protect channels going to amplified speakers

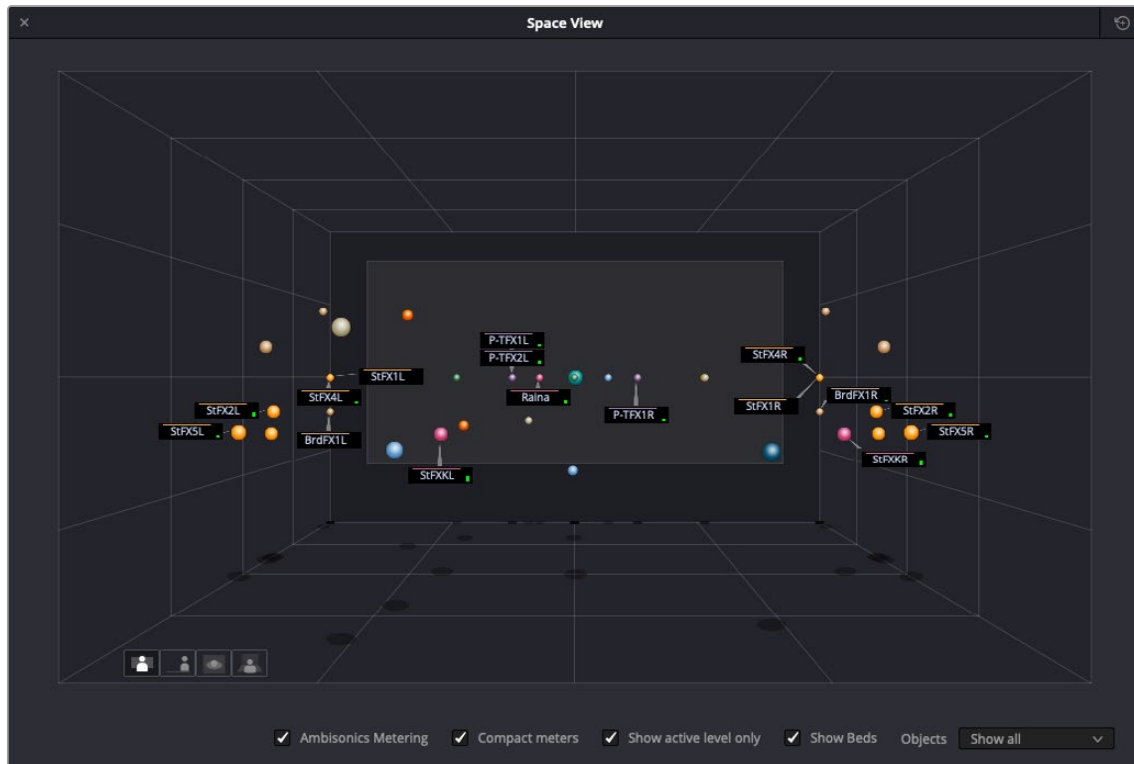
Space View

Space View is a 3D interactive workspace that lets you view your unmuted sound sources' real-time positions and automated movements.

You can freely reposition sources in the 3D sound field according to your mouse movements (Left/Right and Forward/Backward). Holding down Command while moving your mouse forward and backward moves the sound source up and down in the sound field. All your changes can be automated (Touch and Pan) and saved.

Space View can be opened by going to Fairlight > Immersive and selecting Space View.

NOTE: Only unmuted sources with a fader level greater than -70dB will be visible.



Space View of an Immersive mix

Space View Controls

The Space View offers the following control options, which, when used in conjunction with the track mute and solo buttons offer very flexible options that let you focus on individual sources or groups of sources.

In the lower right corner of the workspace, you'll find four buttons that let you view your sound sources from different angles or perspectives.

You can freely rotate the workspace by holding down Command-Option-Shift while dragging it. If you want to get back to a default view, click one of the perspective buttons.

Along the lower edge of the window, you'll find the following checkboxes:

- **Ambisonics Metering:** This checkbox activates Ambisonics sonar-type metering when working on Ambisonic mixes.
- **Compact Meters:** This option replaces the larger halo-style meters with smaller bar-graphs.
- **Show Active Level Only:** Clicking this option hides the names of silent sound sources.
- **Show Beds:** This option reveals the labels and meters for sound sources that are part of sound beds instead of just the signals feeding your master Dolby Atmos bus.

The options in the dropdown menu to the right of the checkboxes determine which objects appear in the virtual space:

- **Show Only Selected:** Shows sources based on tracks you've selected in the Mixer or timeline.
- **Show Visible:** Shows only the active audio sources for tracks that are not hidden.
- **Show All:** All sound sources are visible and labeled.



Deliver

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